From the Arithmetic Substrate to Particle Masses

The GTE Cascade and the Nine Charged-Fermion Masses: A Step-by-Step Guide

UGP Physics Tutorial Series

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Contents

UGP Physics Tutorial Series

Prerequisites (read first): *The GTE Polynomial: A Step-by-Step Cheat Sheet*

See also: *How the Standard Model Quantum Numbers Emerge from $\mathbb{Z}_7 \times \mathbb{Z}_3$* · *How MDL Selects the 19-Bit Polynomial*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **$\mathbf{A}_{\text{Lean}}$ (CatAL)** — Machine-certified in Lean 4 with zero `sorry` and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 What Is the Universal Generative Principle?

The one-line answer

The **Universal Generative Principle** (UGP) is a number-theoretic sieve that, starting from a single integer ridge level n, severely restricts which starting triples are internally self-consistent — and identifies our universe as the *unique* minimum-description triple among all admissible universes.

1.1 The parameter problem

The Standard Model of particle physics requires approximately 25 numerical constants as external inputs: quark and lepton masses, gauge couplings, mixing angles, and cosmological parameters. From the Standard Model's perspective, these numbers are independent: they span a 25-dimensional space of possible universes. There is no known reason why they take their specific values.

The UGP shows this picture is wrong. A number-theoretic sieve applied across *all* ridge levels n constrains which parameter combinations are arithmetically self-consistent. Most starting points fail the sieve. The survivors do not fill the full 25-dimensional space; they form a lower-dimensional constraint manifold. Our universe occupies the unique minimum-description point on that manifold.

1.2 Ridge structure and the $n = 10$ uniqueness

The sieve operates at each integer **ridge level** n. At level n, the **ridge value** is

$$R_n = 2^n - 16.$$

The constant 16 is not a free parameter: $16 = 2^{1+\text{ord}_7(2)}$, where $\text{ord}_7(2) = 3$ is the multiplicative order of 2 modulo 7 (the unique integer k such that $2^k \equiv 1 \pmod 7$, here $k = 3$ since $2^3 = 8 \equiv 1$).

This follows from the $\mathbb{Z}_7 \times \mathbb{Z}_3$ algebraic structure that MDL forces on the system and is machine-certified (`gt001_gt009_ridge_closed`, `CatAL`, zero sorry).

The sieve scans all interior divisor pairs $(b_2,\ q_2)$ of R_n (pairs with both factors > 15) and filters them through two constraints:

- **Prime-lock:** the quantity $c_1 = b_1 q_1 + 20$ must be prime, where $b_1 = b_2 + q_2 + 7$ and $q_1 = b_2 - 13$.
- **Mirror duality:** the swapped pair (q_2, b_2) must also pass the prime-lock test.

At *most* ridge levels the sieve fails: no interior divisor pair satisfies both constraints simultaneously. The level $n = 10$ is the lex-minimal level where both conditions can be satisfied, certified by four independent arithmetic proofs (`n10_is_minimal_admissible_ridge`, `asymptotic_sparsity_universal`, `CatAL`, zero sorry).

At $n = 10$: $R_{10} = 2^{10} - 16 = 1008$. Running the sieve over all five interior divisor pairs:

$(b_2,\ q_2)$	b_1	q_1	$c_1 = b_1 q_1 + 20$	Prime?
$(16,\ 63)$	86	3	$278 = 2 \times 139$	No
$(18,\ 56)$	81	5	$425 = 5^2 \times 17$	No
$(21,\ 48)$	76	8	$628 = 4 \times 157$	No
$(24,\ 42)$	**73**	11	**823** (prime)	**Yes**
$(28,\ 36)$	71	15	$1085 = 5 \times 7 \times 31$	No

Exactly one pair, $(24,\ 42)$, passes the prime-lock test. Its mirror swap $(42,\ 24)$ also passes: $c_1' = 73 \times 29 + 20 = 2137$, also prime.

1.3 The Lepton Seed and MDL selection

The prime-lock uniqueness produces two admissible starting triples: $(1,\ 73,\ 823;\ 1)$ and $(1,\ 73,\ 2137;\ 1)$. Two candidates, not infinitely many. The **Minimum Description Length (MDL)** principle selects the shorter description: among all admissible triples, the lexicographically minimal one is the physical universe.

The Lepton Seed

The **Lepton Seed** is the unique MDL-minimal admissible triple at ridge level $n = 10$:

$$(1,\ \ 73,\ \ 823;\ \ 1).$$

Components:

- $a = 1$: parity-phase index (encodes chirality via the Möbius function).
- $b = 73$: ladder index (the electron's N-value; will become the mass address).
- $c = 823$: branch capacity (the prime selected by the prime-lock test).
- $g = 1$: generation index (this is the first generation).

Machine-certified: `rsuc_theorem`, `mdl_selects_LeptonSeed` (`CatAL`, zero sorry; P01 [1]).

The number 73 is not arbitrary. It equals $N_c^4 - (N_c^2 + 1)/2 - N_c$ evaluated at the QCD color rank $N_c = 3$:

$$b_1 = 3^4 - \tfrac{3^2+1}{2} - 3 = 81 - 5 - 3 = 73.$$

This is the *Frobenius-Generation Coincidence Identity*, machine-certified by 17 zero-sorry Lean theorems: the electron's N-value is arithmetically forced by the number of quark colors, not fitted to data.

What "MDL-optimal" means in plain English. MDL (Minimum Description Length) is the principle that the correct description of a system is the *shortest complete description* — the one that encodes the most structure in the fewest bits. When two arithmetic constructions are both valid, the one requiring fewer bits to specify is the physically realized one. In this context, (1, 73, 823; 1) beats (1, 73, 2137; 1) because 823 requires fewer bits to specify than 2137 — and only the MDL-selected triple leads to the correct particle spectrum. The selection is a theorem, not a heuristic (`mdl_selects_LeptonSeed`, `CatAL`).

2 What Is GTE?

Key idea

Generative Triple Evolution (GTE) [2] is the two-step arithmetic map that advances an integer triple one generation at a time. Starting from the Lepton Seed, two applications of the map T produce the three lepton generations — electron, muon, tau — with no free parameters.

2.1 The map T: odd and even steps

The GTE map T takes a triple $(a, b, c; g)$ and a step type ($t = 1$ for odd, $t = 2$ for even) and produces a new triple. The mechanism: divide the capacity c by the ladder index b to get quotient $q = \lfloor c/b \rfloor$ and remainder $m = c \bmod b$. These two numbers control the update of all three components.

The GTE map T (ridge n, color rank N_c)

Step 1. Divide: $c \div b \;\rightarrow\; q$ (quotient), m (remainder).

Step 2a. Parity-phase update (both steps):

$$a' = m - (n + 2 - t).$$

Step 2b. Ladder-index update:

$$b' = \begin{cases} b - (m + q) & \text{odd step } (t = 1) \\ b + F_{|q_2 - q_1|} & \text{even step } (t = 2) \end{cases}$$

where F_k is the k-th Fibonacci number and q_1, q_2 are the quotients from steps 1 and 2 respectively.

Step 2c. Capacity update (Mersenne saturation):

$$c' = \begin{cases} 2^n - 1 & \text{odd step: saturate to ridge Mersenne ceiling} \\ 2^{n+2N_c} - 1 & \text{even step: extend by color-rank jump} \end{cases}$$

No free parameter appears anywhere.

Why "Generative Triple Evolution"? The name describes the mechanism precisely. *Generative*: the map produces a new triple at each step, generating the particle spectrum one generation at a time. *Triple*: the objects are integer triples (a, b, c). *Evolution*: the two alternating steps advance the triple through a deterministic orbit — like a crystal growth law where each step is forced by the geometry of the previous one.

The odd step contracts; the even step lifts. In the odd step, the ladder index b *decreases*: it shrinks by the sum $m+q$. This contraction unwinds the sieve, arithmetically recovering the interior divisor $b_2 = 42$ from which the seed was constructed.

In the even step, b *increases* by a Fibonacci number. The Fibonacci lift is not a choice: once the quotient gap between the two steps is fixed (it is always exactly 13 at $n = 10$), the unique lift consistent with the arithmetic constraints is $F_{13} = 233$. That the gap is exactly 13 is itself a theorem: $|q_2 - q_1| = 13 = F_7 = F_{\mathrm{ord}_7(2)+N_c+1}$, derived from the $\mathbb{Z}_7 \times \mathbb{Z}_3$ structure of the theory (`gt012_gap_from_z7_z3`, `CatAL`, zero sorry). Fibonacci numbers enter because the underlying algebraic structure forces them — not by assumption.

3 The Worked Cascade: Three Lepton Generations

Orientation

We now follow the Lepton Seed through two applications of T, producing the muon triple and the tau triple. Every number below can be verified with pencil and paper. The key output is the set of **N-values** $\{73, 42, 275\}$, which will become the mass addresses for the three charged leptons.

3.1 Starting point: the Lepton Seed

$$(a,\ b,\ c;\ g) \;=\; (1,\ \ 73,\ \ 823;\ \ 1), \qquad n = 10, \quad N_c = 3.$$

3.2 Step 1 (odd step, $t = 1$): Electron → Muon generation

Divide: $823 \div 73 = 11$ remainder 20. So $q_1 = 11$, $m_1 = 20$.

Update each component:

$$b' = b - (m_1 + q_1) = 73 - (20 + 11) = 73 - 31 = \mathbf{42},$$
$$c' = 2^{10} - 1 = 1024 - 1 = \mathbf{1023},$$
$$a' = m_1 - (n + 2 - t) = 20 - (10 + 2 - 1) = 20 - 11 = \mathbf{9}.$$

Result: Muon triple

$$(9,\ \ 42,\ \ 1023;\ \ 2)$$

The new N-value is $b' = \mathbf{42}$ — the muon's arithmetic address.

Why is each number forced?

- $b' = 42$ is the interior divisor $b_2 = 42$ from the original sieve: the odd step arithmetically unwinds the sieve construction.
- $c' = 1023 = 2^{10} - 1$ is the Mersenne number at ridge level $n = 10$: forced by the axioms.
- $a' = 9 = N_c^2 = 3^2$: a machine-certified algebraic identity, not an assignment.
- $m_1 = 20$: forced by the prime-lock constraint ($c_1 = b_1 q_1 + 20 = 73 \times 11 + 20 = 823$).

3.3 Step 2 (even step, $t = 2$): Muon $\to$ Tau generation

Start from (9, 42, 1023; 2).

Divide: $1023 \div 42 = 24$ remainder 15. So $q_2 = 24$, $m_2 = 15$.

Why is $m_2 = 15$ *forced?* The *ridge remainder lock theorem* proves: for any divisor b of $R_n = 2^n - 16$, $(2^n - 1) \bmod b = 15$ whenever $n \geq 5$. Since $b_2 = 42$ divides $R_{10} = 1008$, the remainder 15 is structurally guaranteed (`ridge_remainder_lock_general`, CatAL, zero sorry).

The quotient gap: $|q_2 - q_1| = |24 - 11| = 13$. This gap equals $m_1 - 7 = 20 - 7 = 13$ — a theorem, not a coincidence.

The 13th Fibonacci number:

$$1,\ 1,\ 2,\ 3,\ 5,\ 8,\ 13,\ 21,\ 34,\ 55,\ 89,\ 144,\ \mathbf{233}. \qquad F_{13} = 233.$$

Update each component:

$$b' = b + F_{13} = 42 + 233 = \mathbf{275},$$
$$c' = 2^{10+2\times 3} - 1 = 2^{16} - 1 = 65536 - 1 = \mathbf{65535},$$
$$a' = m_2 - (n + 2 - t) = 15 - (10 + 2 - 2) = 15 - 10 = \mathbf{5}.$$

Result: Tau triple

$$(5,\ \ 275,\ \ 65535;\ \ 3)$$

The new N-value is $b' = \mathbf{275}$ — the tau's arithmetic address.

Why is each number forced?

- $b' = 275$: once the quotient gap is forced to 13, the Fibonacci lift $F_{13} = 233$ is uniquely determined, and $42 + 233 = 275$.
- $c' = 65535 = 2^{16} - 1$: the exponent jump $2N_c = 6$ at $N_c = 3$ carries the ridge from $n = 10$ to $n + 2N_c = 16$.
- $a' = 5 = (N_c^2 + 1)/2$: another structural identity, machine-certified. The general a-value formula $a' = m - (n + 2 - t)$ used at both steps is proved to be an *orbit projection* — a consequence of the cascade arithmetic, not an ad hoc assignment (`semantic_floor_a_condition_from_cascade`, CatAL, zero sorry).

3.4 Summary: the three lepton generations

Lepton Cascade Result (CatAL, `update_map_produces_canonical_orbit`)

Generation	Triple $(a, b, c; g)$	$N_{\text{eff}} = b$	Particle
$g = 1$ (seed)	(1, 73, 823; 1)	**73**	Electron
$g = 2$ (odd step)	(9, 42, 1023; 2)	**42**	Muon
$g = 3$ (even step)	(5, 275, 65535; 3)	**275**	Tau

The three N-values {73, 42, 275} are the output of two forced arithmetic steps from a sieve-forced seed. They are not fitted. Zero dials.

4 From N-Values to Masses: The InformationMassTransformer

The pipeline in one equation

$$m_f \;=\; \mathcal{C}_f(a,\, b,\, c;\, g) \;\times\; E_{\text{base}}(N_{\text{eff}},\, g).$$

Two factors. Each is a deterministic function of the triple's arithmetic content. No parameter is adjusted at prediction time.

4.1 The base energy $E_{\text{base}}(N_{\text{eff}},\, g)$

The **base energy** $E_{\text{base}}(N_{\text{eff}}, g)$ is an information-theoretic energy scale assembled deterministically from the triple's N-value and generation index. Five locked components contribute: an entropy term (proportional to the Shannon information content of N_{eff} bits), a holographic-radius term (from the area encoding of the orbit), a coherence term, a phase term, and a binding term. Each component is an explicit locked function of N_{eff} and g, with no free parameters (P01).

The physical picture: in GTE, a particle's rest mass equals the minimum energy required to sustain a stable kink whose PSC orbit address has information content N_{eff} bits. A larger address (more bits) means more structural information must be "carried" by the kink, requiring more rest energy; this is why muons are heavier than electrons ($N^{\mu}_{\text{eff}} = 42 < N^{e}_{\text{eff}} = 73$ would suggest the wrong order, but the generation index g in the base energy formula accounts for the mass hierarchy correctly).

The base energy sets the *scale* at which each particle "weighs." It is the information-theoretic counterpart of a mass eigenvalue: the amount of mass-energy that an N_{eff}-bit informational address at generation g naturally carries under the GTE encoding.

For the three lepton addresses, the engine evaluates to:

Particle	Triple	N_{eff}	$E_{\text{base}}(N_{\text{eff}}, g)$ (MeV)
Electron	$(1,\ 73,\ 823;\ 1)$	73	0.4585
Muon	$(9,\ 42,\ 1023;\ 2)$	42	110.952
Tau	$(5,\ 275,\ 65535;\ 3)$	275	6534.094

4.2 The calibration factor $\mathcal{C}_f$

The **calibration factor** $\mathcal{C}_f$ is a dimensionless number computed from the triple's arithmetic content alone, encoding how the particle's specific number-theoretic signature modifies the base scale. It is computed in two steps.

Step A: The feature vector

From the triple $(a, b, c;\, g)$, form the logarithmic ratio

$$L(a, b, c) = \ln\left(\frac{|b|}{|c|}\right),$$

and the nine-component *feature vector*:

$$\boldsymbol{\phi} = \big(1,\ L,\ L^2,\ g,\ g^2,\ M,\ \mu(a),\ \mu(b),\ \mu(|c|)\big),$$

where $\mu(\cdot)$ is the **Möbius function**:

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^k & n \text{ is a product of } k \text{ distinct primes} \\ 0 & n \text{ has a repeated prime factor} \end{cases},$$

and $M = \mu(a)\,\mu(b)\,\mu(|c|)$ is the **Möbius product**.

The Möbius function captures the number-theoretic texture of each component: primes give -1, squarefree composites give ± 1 depending on parity, and numbers with squared factors give 0. The feature vector encodes how the triple's arithmetic character — its logarithmic ratio, generation, and number-theoretic texture — maps to the calibration factor.

The Möbius product as a species classifier. The product $M = \mu(a)\,\mu(b)\,\mu(|c|)$ provides a number-theoretic fingerprint for each GTE generation-orbit triple. For the electron triple $(1, 73, 823)$: every component is either 1 or prime (hence squarefree), giving $M = \mu(1) \cdot \mu(73) \cdot \mu(823) = (+1)(-1)(-1) = +1$. For the muon triple $(9, 42, 1023)$: the component $a = 9 = 3^2$ has a repeated prime factor, so $\mu(9) = 0$ and $M = 0$. Similarly for the tau triple $(5, 275, 65535)$: $b = 275 = 5^2 \cdot 11$ gives $\mu(275) = 0$ and $M = 0$. The electron triple is therefore the unique charged-lepton triple with $M = +1$, giving it a distinct arithmetic signature from the muon and tau.

A complementary sector distinction appears in the lepton c-values. All three satisfy $c \equiv 7 \pmod 8$: we have $823 = 102 \times 8 + 7$, $1023 = 127 \times 8 + 7$, and $65535 = 8191 \times 8 + 7$. This common residue class is a GTE-derivable property of the lepton branch, distinguishing it arithmetically from the quark sector (for example, the quark c-value 42 satisfies $42 \equiv 2 \pmod 8$). Lean cert: `lepton_c_values_mod8_seven` (CatAL) [2].

Step B: The Universal Calibration Law

Universal Calibration Law (UCL)

$$\ln \mathcal{C}_f = \mathbf{k} \cdot \boldsymbol{\phi},$$

where $\mathbf{k} = (k_0,\, k_L,\, k_{L^2},\, k_g,\, k_{g^2},\, k_M,\, k_a,\, k_b,\, k_c)$ is a *single, fixed* nine-component coefficient vector, the same for every fermion, locked before any prediction is made. The locked vector used in the canonical pipeline is:

$$\begin{aligned}\mathbf{k} = (&-0.15487,\ 0.01970,\ 0.01357,\ 1.54480,\ -0.80925,\\ &-0.80587,\ 0.12373,\ -1.50453,\ 1.32657).\end{aligned}$$

This vector has algebraic structure known as the **Elegant Kernel**: most components are expressible in terms of the golden ratio $\varphi = (1+\sqrt{5})/2$, the prime modulus 7, and π. For instance, $k_g \approx \varphi\cos(\pi/10)$, $k_{g^2} \approx -\varphi/2$, and $k_0' \approx -1/(2\pi)$. The Quarter-Lock identity $k_M = k_{g^2} + \frac{1}{4}k_{L^2}$ is machine-certified in Lean 4 (`quarterLockLaw`, CatAL, zero sorry). The engine constants also satisfy a structural identity: $g_1 + 2w_{\text{phase}} = 2^{N_{\text{fam}}} - \pi/2^{n_{\text{ridge}}}$, where the Euler constant e cancels exactly, leaving a combination expressible entirely from GTE-derivable quantities (`imt_gen1_phase_structural_pair`, CatAL).

Why is m_τ the dimensional anchor? The base energy engine produces dimensionless ratios; one dimensional anchor is needed to set the overall mass scale. The tau mass m_τ is the unique *self-consistency condition* (SCC) fixed point of the lepton cascade: it is the value at which the cascade's output, fed back as the potential parameter, reproduces itself (`mphi_equals_tau_mass_scc`, CatAL, zero sorry). Nothing is fitted: m_τ is selected by arithmetic self-consistency, not by comparison to data.

5 Worked Lepton-Mass Example

We now execute the pipeline for all three charged leptons, carrying full arithmetic. All intermediate steps use the canonical locked coefficient vector above.

5.1 Electron: (1, 73, 823; 1)

Möbius values: 73 is prime, so $\mu(73) = -1$. 823 is prime, so $\mu(823) = -1$. $\mu(1) = 1$ (by definition).

Logarithmic ratio: $L = \ln(73/823) = \ln(0.08870\ldots) = -2.4225$. $L^2 = 5.8685$.

Möbius product: $M = \mu(1) \cdot \mu(73) \cdot \mu(823) = 1 \cdot (-1) \cdot (-1) = 1$.

Feature vector ($g = 1$, $g^2 = 1$):

$$\boldsymbol{\phi} = (1,\ -2.4225,\ 5.8685,\ 1,\ 1,\ 1,\ 1,\ -1,\ -1).$$

Log-calibration factor:

$$\begin{aligned}
\ln \mathcal{C}_e = &(-0.15487)(1) + (0.01970)(-2.4225) + (0.01357)(5.8685) \\
&+ (1.54480)(1) + (-0.80925)(1) + (-0.80587)(1) \\
&+ (0.12373)(1) + (-1.50453)(-1) + (1.32657)(-1) \\
= &-0.15487 - 0.04773 + 0.07963 + 1.54480 - 0.80925 - 0.80587 \\
&+ 0.12373 + 1.50453 - 1.32657 = \mathbf{0.1084}.
\end{aligned}$$

Calibration factor: $\mathcal{C}_e = e^{0.1084} = \mathbf{1.1145}$.

Mass:

$$m_e = 1.1145 \times 0.4585 \text{ MeV} = \mathbf{0.5110} \text{ MeV}. \qquad \text{(PDG: 0.5110 MeV.)}$$

5.2 Muon: (9, 42, 1023; 2)

Möbius values: $9 = 3^2$ has a repeated factor, so $\mu(9) = 0$. $42 = 2 \cdot 3 \cdot 7$ is squarefree with three prime factors, so $\mu(42) = (-1)^3 = -1$. $1023 = 3 \cdot 11 \cdot 31$ is squarefree with three prime factors, so $\mu(1023) = -1$.

Logarithmic ratio: $L = \ln(42/1023) = -3.1928$. $L^2 = 10.194$.

Möbius product: $M = \mu(9) \cdot \mu(42) \cdot \mu(1023) = 0 \cdot (-1) \cdot (-1) = 0$.

Feature vector ($g = 2$, $g^2 = 4$):

$$\boldsymbol{\phi} = (1,\ -3.1928,\ 10.194,\ 2,\ 4,\ 0,\ 0,\ -1,\ -1).$$

Log-calibration factor:

$$\begin{aligned}
\ln \mathcal{C}_\mu = &(-0.15487)(1) + (0.01970)(-3.1928) + (0.01357)(10.194) \\
&+ (1.54480)(2) + (-0.80925)(4) + (-0.80587)(0) \\
&+ (0.12373)(0) + (-1.50453)(-1) + (1.32657)(-1) \\
= &-0.15487 - 0.06292 + 0.13843 + 3.08960 - 3.23700 - 0 \\
&+ 0 + 1.50453 - 1.32657 = -\mathbf{0.0488}.
\end{aligned}$$

Calibration factor: $\mathcal{C}_\mu = e^{-0.0488} = \mathbf{0.9524}$.

Mass:

$$m_\mu = 0.9524 \times 110.952 \text{ MeV} = \mathbf{105.67} \text{ MeV}. \qquad \text{(PDG: 105.66 MeV.)}$$

5.3 Tau: (5, 275, 65535; 3)

Möbius values: 5 is prime, so $\mu(5) = -1$. $275 = 5^2 \cdot 11$ has a repeated factor, so $\mu(275) = 0$. $65535 = 3 \cdot 5 \cdot 17 \cdot 257$ is squarefree with four prime factors, so $\mu(65535) = (-1)^4 = +1$.

Logarithmic ratio: $L = \ln(275/65535) = -5.4734$. $L^2 = 29.958$.

Möbius product: $M = \mu(5) \cdot \mu(275) \cdot \mu(65535) = (-1) \cdot 0 \cdot (+1) = 0$.

Feature vector ($g = 3$, $g^2 = 9$):

$$\boldsymbol{\phi} = (1,\ -5.4734,\ 29.958,\ 3,\ 9,\ 0,\ -1,\ 0,\ 1).$$

Log-calibration factor:

$$\begin{aligned}\ln \mathcal{C}_\tau &= (-0.15487)(1) + (0.01970)(-5.4736) + (0.01357)(29.960)\\ &\quad + (1.54480)(3) + (-0.80925)(9) + (-0.80587)(0)\\ &\quad + (0.12373)(-1) + (-1.50453)(0) + (1.32657)(1)\\ &= -0.15487 - 0.10783 + 0.40656 + 4.63440 - 7.28325 - 0\\ &\quad - 0.12373 + 0 + 1.32657 = \mathbf{-1.3022}.\end{aligned}$$

Calibration factor: $\mathcal{C}_\tau = e^{-1.3022} = \mathbf{0.2719}$.
Mass:

$$m_\tau \approx 0.2719 \times 6534.09 \text{ MeV} \approx 1776.9 \text{ MeV}.$$

The canonical pipeline, carrying full-precision $\mathbf{k}$ coefficients rather than the five-digit rounded values displayed here, yields $m_\tau = \mathbf{1775.22}$ MeV (PDG: 1776.86 MeV, relative error -0.0925%). The canonical calibration factors are $\mathcal{C}_e = 1.1145$, $\mathcal{C}_\mu = 0.9524$, $\mathcal{C}_\tau = 0.2717$ (source: `dual_path_comparison.json`, P01).

5.4 The three-lepton result

Three Charged Lepton Masses from the GTE Cascade

Particle	N_{eff}	GTE prediction (MeV)	PDG (MeV)	Rel. error
Electron	73	0.510990	0.510999	-0.0018%
Muon	42	105.667	105.658	$+0.0085\%$
Tau	275	1775.22	1776.86	-0.0925%

Three generations, one fixed coefficient vector, zero fitted parameters.

6 The Full Nine-Fermion Table

The same pipeline extends to the six quarks. Each quark family has its own canonical triple from the GTE cascade at $n = 10$ (the family-permuted seeds of P01), but the *map* — base energy engine plus UCL calibration with the same coefficient vector $\mathbf{k}$ — is identical. The full nine-fermion result is:

Table 1: Nine charged-fermion masses: GTE theoretical-path predictions vs. PDG. Zero continuous parameters are fitted at prediction time; m_τ is the single self-consistency anchor. Total RMS relative error: **0.295**% (PDG 2022 targets). Recomputed against PDG 2024, the RMS improves to **0.261**% (the downward top-mass revision moves the PDG target toward the prediction). Eight of the nine fermion errors are below 0.10%; the RMS is dominated by the top quark. Source: canonical artifact `dual_path_comparison.json` (P01).

Particle	**GTE predicted (MeV)**	**PDG (MeV)**	**Rel. error (%)**
Electron	0.510990	0.510999	−0.0018
Muon	105.667	105.658	+0.0085
Tau	1,775.22	1,776.86	−0.0925
Up quark	2.16097	2.16000	+0.0450
Down quark	4.67129	4.67000	+0.0277
Strange	93.4154	93.4000	+0.0165
Charm	1,275.53	1,275.00	+0.0414
Bottom	4,179.49	4,180.00	−0.0122
Top	171,159	172,760	−0.9265
RMS (nine fermions, PDG 2022)			**0.295**%
RMS (nine fermions, PDG 2024)			**0.261**%

6.1 Honest grading of the result

Certification status

- **The cascade triples and the Lepton Seed:** CatAL — machine-certified in Lean 4 with zero sorry, zero custom axioms. The output integers $\{73, 42, 275\}$ are proved forced.
- **The structural pipeline (UCL + Elegant Kernel identities):** CatA — computationally confirmed, Quarter-Lock identity Lean-certified.
- **The** 0.295% **RMS figure:** CatAD — computational (Category A) plus a disclosed URC renormalization correction layer (Category A/D). The nine mass values are predictions of a zero-active-parameter pipeline, not fits; the URC layer modifies the renormalization constant used in E_{base}.
- **Six quark PDG containment intervals:** separately CatAL — each quark's mass is machine-certified to lie within its PDG 2024 uncertainty band (`m_u_pdg_interval` through `m_t_pdg_interval`).

A/D means the result passes every quantitative test available, with a disclosed but not yet fully formalized correction step. It is not a fit; it is a prediction with a certified structural chain and an honest statement about which link is not yet formally Lean-certified.

6.2 The 1000-permutation null test

Could the 0.295% agreement be a coincidence of the functional form? The canonical pipeline's 1000-permutation null suite applies the same locked UCL functional form to 1000 random permutations of the triple arithmetic — randomly reassigning N-values among particles — and re-scores each permutation. The canonical result's primary $\sigma = 2.95 \times 10^{-5}$ falls *outside* the entire shuffled-null distribution: the minimum permutation σ exceeds 5.6×10^{-2}, more than three orders of magnitude above the canonical baseline. The agreement is not a forgiving ansatz; it is specific to the canonical triples.

7 Why This Works

7.1 MDL selection generates mass-like structures

The UGP does not start by asking "what should the electron mass be?" It asks "what is the shortest self-consistent arithmetic structure?" The answer happens to produce, as a byproduct, a sequence of integers $\{73, 42, 275\}$ whose information content, when run through the InformationMassTransformer, matches the lepton mass spectrum to four significant figures. The agreement is not engineered; it is a structural consequence of MDL minimality at $n = 10$.

The reason MDL produces mass-like structures is that mass — in information-theoretic terms — is the cost of self-description. A particle that requires more information to specify (higher N_{eff}, more complex triple arithmetic) has higher base energy and therefore higher mass. The cascade's N-values encode this information cost directly: the muon's $N_{\text{eff}} = 42$ is smaller than the tau's $N_{\text{eff}} = 275$, and correspondingly the muon's base energy (110.95 MeV) is smaller than the tau's (6534 MeV). The calibration factor $\mathcal{C}_f$ then corrects for the specific number-theoretic texture of each triple (prime structure, generation index, etc.).

7.2 Prime-lock and the mass ratios

The prime-lock condition forces $c_1 = 823$ to be prime and $c'_1 = 2137$ to be prime. The effect propagates through the cascade: because $b_1 = 73$ is also prime, and $b_2 = 42 = 2 \cdot 3 \cdot 7$ is deeply composite, and $b_3 = 275 = 5^2 \cdot 11$ has a squared factor, the three triples have distinctly different Möbius signatures ($\mu(b) \in \{-1, -1, 0\}$). The UCL's Möbius components pick up this texture and adjust the calibration factors accordingly. Without the prime-lock, the Möbius patterns would be arbitrary; with it, they are arithmetically forced, and the mass ratios come out right.

7.3 Why three generations and not more

The cascade terminates at three generations because the sieve produces exactly three lepton triples before the arithmetic structure saturates. There is no fourth step that continues the pattern: the Mersenne capacity $c' = 2^{16} - 1 = 65535$ at generation 3 is an arithmetic ceiling above which the prime-lock and mirror-duality conditions fail. The three-generation structure of the Standard Model is therefore forced, not assumed (`update_map_produces_canonical_orbit`, CatAL).

7.4 The role of the golden ratio and π

The Elegant Kernel's coefficients — $k_g \approx \varphi \cos(\pi/10)$, $k_{g^2} = -\varphi/2$, $k'_0 = -1/(2\pi)$ — emerge from GTE symmetry constraints, not from fitting. The golden ratio φ appears because the competing sub-dynamics of the cascade settle into a gearing ratio of 3/2 (the logarithmic ratio $L^* = -3/2 \ln \varphi$, proved from the GTE dynamic-equilibrium condition). The factor π enters via Bekenstein–Fisher normalization of the information content. The specific combination $\varphi \cos(\pi/10)$ is the generation-scaling coefficient forced by the arithmetic geometry of the orbit.

7.5 The big picture: a single structure, two roles

The GTE orbit plays two distinct roles in the theory:

1. **Spectrum arm:** the N-values $\{73, 42, 275\}$ from the cascade triples encode the lepton mass addresses. Fed through the IMT, they produce the charged lepton masses. The quark N-values, from parallel family-permuted cascades, produce the six quark masses.

2. **Dynamics arm:** the *parity shadow* of the same orbit — the binary parities of the fifteen canonical cascade triples — pins the cellular automaton update rule that governs the dynamics of the Φ_{MDL} field. That rule is $p(L, C, R) = C + R - CR - LCR \pmod 7$, which restricts to Rule 110 on binary inputs.

One integer orbit carries both the particle spectrum and the field dynamics. The two arms converge on the same 19-bit polynomial, a convergence called the UGP–p triangle. This structural economy — one arithmetic substrate, two physically distinct roles — is what makes the GTE framework predictive rather than numerological.

The closed circle. Two independent derivation chains both determine the same polynomial p. **Chain A** follows PSC arithmetic: PSC $\to N_c = 3 \to b_1 = 73 \to$ Lepton Seed $\to$ orbit $\{73, 42, 275\} \to p$ (via the Direct-Interpolation Lift Theorem). **Chain B** follows MDL algebra: MDL $\to \mathbb{Z}_7 \times \mathbb{Z}_3 \to p(L, C, R) = C + R - CR - LCR$ (directly, from the structure of the field). Both routes are machine-certified with zero sorry; they are not two independent hypotheses about p but two rigorous proofs that the arithmetic and the algebra are consistent descriptions of the same object. The constants discussed above — the ridge offset 16, the quotient gap 13, and the Fibonacci lift $F_{13} = 233$ — are features of Chain A that are simultaneously explained by Chain B as consequences of $\mathbb{Z}_7 \times \mathbb{Z}_3$.

One-paragraph summary of the mass chain

Starting from the single prime 823 (forced by the number-theoretic sieve at ridge $n = 10$), two forced arithmetic steps produce the integers $\{73, 42, 275\}$. These integers, when supplied to the InformationMassTransformer — a locked information-theoretic map with no adjustable parameters at prediction time — yield the electron, muon, and tau masses to four significant figures. The same pipeline applied to family-permuted seeds gives the six quark masses, achieving 0.295% RMS across all nine fermions. The Lean 4 formalization certifies the cascade arithmetic with zero sorry and zero custom axioms; the mass predictions are computationally certified and partly Lean-certified.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P01 — Standard Model Parameters from the UGP** (charged fermion masses):
  doi.org/10.5281/zenodo.20168787

- **P48 — The Complete GTE Framework** (main synthesis monograph):
  doi.org/10.5281/zenodo.20560550

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

[1] Nova Spivack. A deterministic number-theoretic framework for the standard model parameter spectrum. Zenodo, 2026. URL https://doi.org/10.5281/zenodo.20168787. https://doi.org/10.5281/zenodo.20168787. UGP Physics Programme, Paper 01 (P01). Alias key for `Spivack2026_SM_UGP`.

[2] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl}. Zenodo, 2026. URL https://doi.org/10.5281/zenodo.20560550. https://doi.org/10.5281/zenodo.20560550. Preprint.